Interpretable and Uncertainty-Aware Crop Price Forecasting for Indian Markets using Temporal Fusion Transformers

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***Abstract*—Monthly retail food prices in India are difficult to forecast. A single number is rarely enough for a trader or a procurement officer to act on. A useful forecast must also give a range and some idea of what drives the prediction. This paper presents a system for three crops — onions, tomatoes and rice — that outputs a median forecast, a calibrated band, and a per-feature explanation for each of the next one to six months. Price data is taken from the WFP India retail panel. Weather data is taken from the NASA POWER monthly API.**

XGBoost is used as the point baseline. A Temporal Fusion Transformer (TFT) is used for quantile forecasting. The XGBoost output is added as one of the known future inputs to TFT. Conformal Quantile Regression (CQR) then adjusts the $[q_{0.1}, q_{0.9}]$ band separately for each crop. The 2022 window is used as the calibration set.

On 792 test rows from January 2023 onward, the fused model gives 11.4 percent MAPE and 84.0 percent empirical band coverage. The paired t-test against a plain TFT gives $p = 4.6 \times 10^{-86}$, and Cohen's d is -0.79. The Variable Selection Network ranking is stable across 1000 bootstrap resamples (Kendall $\tau = 0.942$). All artifacts are served through a Streamlit dashboard.

***Index Terms*—crop price forecasting, Temporal Fusion Transformer, probabilistic forecasting, conformal calibration, uncertainty quantification, agricultural market analytics**

I. INTRODUCTION

Food crop prices in India shift fast. Onion prices can move from around per kg Rs. 30 to Rs. 60 in a few months when the monsoon is late. Tomato prices in July 2023 climbed even higher. Such moves hit household food budgets, show up in retail inflation, and force the Food Corporation of India (FCI) to take buying and storage calls under real risk. In such a setting, a single forecast number is of limited use. A range of likely prices, with some idea of what drives the range, is far more useful.

Studies on Indian crop price forecasting have grown over the past decade. Nayak et al. [1] apply deep models with weather inputs to tomato, onion and potato prices. Manogna et al. [2] compare LSTM, bidirectional RNN and gradient boosted tree models across many commodities. Paul et al. [3] work on brinjal prices. Two gaps are visible across these papers. The first is that every result is reported as a single point error such as MAPE or RMSE. The second is that none of these models exposes which input feature drove a particular prediction, which makes it hard to audit the forecast during a price spike.

The system in this paper closes both gaps for three crops. XGBoost is trained on the engineered lag, rolling and weather features and gives a strong point estimate. A Temporal Fusion Transformer (TFT) is trained on the same data and gives three quantile outputs per month: $q_{0.1}$, $q_{0.5}$ and $q_{0.9}$. The XGBoost output is added as one of the known future inputs to TFT, so the deep model can use it or discount it as required. Conformal Quantile Regression (CQR) then adjusts the band separately for each crop, using 2022 as the calibration year. The Variable Selection Network (VSN) weights from TFT are extracted and bootstrapped to show which features are consistently important.

On the January 2023 onward test period, the fused model reaches 11.4 percent MAPE and 84.0 percent empirical coverage across 792 test rows. The gain over a plain TFT is confirmed by a paired t-test ($p = 4.6 \times 10^{-86}$), a Wilcoxon test ($p = 1.2 \times 10^{-95}$), and a Cohen's d of -0.79.

The rest of the paper is laid out as follows. Section II states the contributions. Section III reviews related work by model family. Section IV describes the dataset. Section V covers the feature engineering. Section VI describes the pipeline. Section VII reports the results with statistical validation. Section VIII describes the dashboard. Section IX closes with limitations and future work.

II. CONTRIBUTIONS

Four design choices separate this work from the reviewed studies.

1. XGBoost output used as a TFT input. An XGBoost model is trained on data up to 2019 only. Its log-price predictions are then computed for the full timeline and attached as a known future input to TFT. TFT's Variable Selection Network decides at each time step how much weight to give this signal. This is not the same as model stacking [4]. In stacking, two models predict independently and the outputs are combined afterward. Here the tree-model output is just one more feature inside TFT. For a stable crop like rice the XGBoost signal is reliable and TFT weights it heavily. For tomatoes during a heat spike, TFT can downweight it and rely on its own sequential context. The 2019 cutoff ensures that the auxiliary model has not seen any 2022 or 2023 data, so the calibration window stays clean.

2. CQR applied per crop on log-price residuals. Conformal Quantile Regression [5] is applied separately for each crop on the 2022 validation window, in log-price space. A single global offset does not work for this dataset. The offset that calibrates rice well leaves the tomato band too narrow, because tomato prices can double in a single month while rice prices barely move.

3. Three statistical tests of every claim. The accuracy gain over the plain TFT is checked with a paired t-test, a Wilcoxon signed-rank test and a Cohen's d effect size on $n = 792$ matched test rows. The first checks the mean difference. The second is a non-parametric backup. The third confirms the effect is large in practice and not just statistically significant from sample size alone. Empirical coverage is checked against the nominal 80 percent target with an exact binomial test, separately per crop. The VSN feature ranking is checked across 1000 bootstrap resamples with pairwise Kendall τ.

4. A working dashboard. The trained artifacts load into a Streamlit application with four views: price history with the model overlay, a six-month forward forecast with the calibrated band and a risk label, an explainability view with VSN and attention weights, and a test-validation view on 2023 data. A non-technical user can run the full workflow without touching the scripts.

III. Literature Review

Prior work on Indian crop price forecasting falls into four groups by model family.

A. Statistical and Classical Methods

ARIMA and SARIMA have been the default baseline for monthly agricultural prices for decades [6]. Many Indian studies have applied them to tomato, onion and rice series because they are simple to fit and their assumptions are easy to state. The structural problem is that these models assume the series is stationary after differencing. Non-linear price spikes such as the late-2019 onion crisis or the mid-2023 tomato surge are not well captured by a model that expects the series to revert. Adding rainfall or temperature as an external regressor requires manual setup, and the lag structure cannot be learned from data.

B. Machine Learning Methods

Random Forest, XGBoost and Support Vector Regression perform well on tabular features such as lags and rolling weather columns, and they routinely beat ARIMA on monthly price data. Manogna et al. [2] show this across many Indian commodities. The limitation is not accuracy. It is scope. The output is one number. There is no native way to attach a band, and feature importance scores such as SHAP or gain are computed globally over the training set. They cannot show which feature was active on the specific month when prices spiked.

C. Recurrent Deep Learning Methods

LSTM, GRU and bidirectional RNN architectures handle sequential dependence better than tabular methods and accept exogenous inputs naturally. Paul et al. [3] and Nayak et al. [1] show competitive results on Indian crop prices with these architectures. Producing a calibrated uncertainty band on top usually requires re-training with a quantile loss or adding Monte Carlo dropout, and the resulting interval does not come with a coverage guarantee.

D. Transformer-Based and Attention Methods

Transformer-based forecasters have grown quickly. DeepAR [7] outputs probabilistic forecasts through learned distributions. Informer [8], N-BEATS [9], NHITS [10] and PatchTST [11] address long horizons through different attention or decomposition designs. The Temporal Fusion Transformer (TFT) [12] is used in this work because it was designed to combine sequential modeling, multi-horizon quantile output, mixed static and time-varying inputs and an interpretability mechanism in the Variable Selection Network. The M4 and M5 competitions [13]–[15] and the analysis by Zeng et al. [16] suggest that no single architecture dominates. The right choice depends on dataset size, update frequency and whether interpretability is required.

The gap specific to Indian crop price forecasting is that no reviewed study produces all of: a calibrated band with empirically tested coverage, a feature ranking validated against resampling noise, and a working interface. Most studies report a single point-error number and stop there. In the present work, the tree model's strength on tabular features is preserved by feeding XGBoost output back into TFT as a known input, and CQR is applied per crop so that the band stays reliable. Table I summarises this positioning.

IV. Dataset

A. Data Sources

Three public sources of Indian crop price data were evaluated. Agmarknet, run by the Government of India, has daily mandi-level wholesale prices for many crops. However, market coverage is patchy across states, and a wholesale-to-retail mapping would introduce extra error. The Department of Consumer Affairs daily retail panel is closer to the target, but the historical depth is short — a few years for most cities — which is not enough for a model with a 24-month encoder window. The WFP India retail dataset [17] was chosen instead. It has monthly retail prices from 1994 onward across more than a hundred markets, with a consistent schema across years. The monthly granularity matches realistic planning horizons. The long history covers the 2008 food price shock, the 2014–15 onion spike and the COVID disruption, all in the training window.

TABLE I
SUMMARY OF LITERATURE GROUPS AND WHAT EACH LEAVES OPEN. "CALIBRATED BAND" MEANS THE PREDICTION INTERVAL HAS BEEN EMPIRICALLY TESTED FOR COVERAGE. "STABLE EXPLANATION" MEANS THE FEATURE RANKING HAS BEEN VALIDATED ACROSS RESAMPLES.

Method group	Point accuracy	Calibrated band	Stable explanation	Deployed dashboard
Statistical (ARIMA, SARIMA)	moderate	no	no	no
ML (RF, XGBoost, SVR)	strong	no	partial (static)	rare
Recurrent (LSTM, GRU)	strong	rare	no	rare
Transformer (DeepAR, TFT, Informer)	strong	sometimes	partial	rare
This work	**strong**	**yes (tested)**	**yes (bootstrap)**	**yes**

For weather, Indian ground station data has gaps in rural market locations. The NASA POWER monthly point API [18] gives satellite-derived monthly estimates at any latitude-longitude pair with no missing values. This makes it practical to match weather to all 53 markets in the panel. Three variables are used: monthly total rainfall, mean temperature and mean relative humidity. These three directly affect vegetable yield during growing and transport.

B. Dataset Preparation

The raw WFP file has 145,124 rows covering 41 commodities, 170 markets and the years 1994 onward. This is filtered down in a few steps to reach the panel used for modeling. Each step in Table II corresponds to one constraint imposed by the modeling pipeline: keeping only retail records (the wholesale series is a different formation process), keeping only KG units (so the price target is comparable across rows), dropping aggregate national records (they are not real market locations), keeping only the three commodities of interest, dropping series shorter than 60 months (TFT needs 24 encoder months plus several training windows), and forward-filling short internal gaps to keep the monthly grid regular.

The resulting panel has 16,939 monthly observations across 123 market-crop series in 53 markets, covering January 1995 to July 2023.

C. Crop Selection

The three crops were picked to span a wide range of price behaviour. Volatility is measured here by the Coefficient of Variation (CV): standard deviation divided by mean, expressed as a percentage.

- **Onions** have a CV of about 60 percent. The kharif and rabi cycles and storage losses make prices volatile. The price episode of late 2019, when retail onion prices climbed sharply, sits in the training window.
- **Tomatoes** have a CV of about 90 percent and are the most volatile of the three. Prices are very sensitive to short-duration heat and rain events. The mid-2023 price episode sits in the test period.
- **Rice** has a CV of about 20 percent and is the most stable, supported by government procurement and the public distribution system. Rice acts as a low-volatility reference: a model that does well on rice but poorly on tomatoes is mostly picking up easy patterns.

D. Temporal Split

A strict chronological split is used throughout so that no future data leaks into training. Table III shows the calendar split. A random split would let 2023 months appear in the training set and inflate the test accuracy, which a deployed forecaster cannot do.

TABLE III
TIME-BASED SPLIT USED FOR THE FINAL MODEL. THE 2022 WINDOW DOUBLES AS THE CALIBRATION SET FOR CQR. THE TEST PERIOD IS THE HELD-OUT 2023 ONWARD DATA THAT THE MODEL NEVER SEES DURING TRAINING OR CALIBRATION.

Split	Date range	Rows	Series
Train	up to 31 Dec 2021	14,686	123
Validation–calibration	1 Jan – 31 Dec 2022	1,461	123
Test	1 Jan 2023 onward	792	123

V. FEATURE ENGINEERING

Index s denotes one market-crop series (e.g., onions in Bangalore, rice in Hyderabad) and t is the month. $p_{s,t}$ is the retail price in Rs. per kg. All features below are computed from the same raw columns and shared by both XGBoost and TFT.

A. Log transform on the target price

$$y_{s,t} = \log(1 + p_{s,t}) \tag{1}$$

The $+1$ shift handles zero-price rows. In log space a tomato spike from per kg Rs. 40 to Rs. 120 has the same magnitude as a move from Rs. 10 to Rs. 30 — the model is not disproportionately pulled by high-price months. Inverse transform to recover price scale:

$$\hat{p}_{s,t} = \exp(\hat{y}_{s,t}) - 1 \tag{2}$$

B. Lag features

$$\mathrm{lag}_1(s,t) = p_{s,t-1} \tag{3}$$
$$\mathrm{lag}_{12}(s,t) = p_{s,t-12} \tag{4}$$

The one-month lag carries short-run momentum. The twelve-month lag gives the model a direct look at the same seasonal window from the previous year, which is particularly useful for onions where kharif and rabi harvest timing repeats annually.

TABLE II

STEP-BY-STEP FILTERING OF THE RAW WFP INDIA DATASET. EACH ROW REMOVES ONE TYPE OF RECORD AND STATES WHY THAT REMOVAL IS NEEDED FOR THE MODELING PIPELINE.

#	Filter step	Rows after	Why the step is needed
0	Raw WFP India price data	145,124	starting point before any filtering
1	Keep retail records only	90,312	wholesale and farm-gate prices follow a different formation process and should not be mixed with retail
2	Keep KG unit only	87,501	a few rows use liters or 100g; mixing units would corrupt the price target
3	Drop national and aggregate records	84,920	rows tagged "National Average" and similar are not real market locations and break the per-market structure
4	Keep onions, tomatoes, rice only	21,458	three commodities chosen to span low, medium and high price volatility (Section IV-C)
5	Drop series with <60 months	17,812	TFT needs at least 24 encoder months plus enough training windows; very short series cannot be used
6	Forward-fill short internal gaps	16,939	keeps a regular monthly panel without imputing across long missing runs

C. Rolling means

$$\text{roll}_3(s,t) = \tfrac{1}{3}\textstyle\sum_{k=0}^{2} p_{s,t-k} \tag{5}$$
$$\text{roll}_6(s,t) = \tfrac{1}{6}\textstyle\sum_{k=0}^{5} p_{s,t-k} \tag{6}$$

The three-month window captures the recent trend without overreacting to a single outlier month. The six-month window gives a slower-moving baseline, so the model can distinguish a brief spike from a genuine regime change in the local price level.

D. Year-on-year change

$$\text{yoy\_change}(s,t) = \frac{p_{s,t} - p_{s,t-12}}{p_{s,t-12}} \tag{7}$$

This fraction tells the model whether the current month is running ahead of or behind the same seasonal window last year, which is more informative than the raw lag value alone.

E. Cyclical month encoding

$$m_{\sin}(t) = \sin\!\left(\tfrac{2\pi\,\text{month}(t)}{12}\right) \tag{8}$$
$$m_{\cos}(t) = \cos\!\left(\tfrac{2\pi\,\text{month}(t)}{12}\right) \tag{9}$$

Integer month encoding creates a false discontinuity between December (12) and January (1), which are in fact adjacent in the annual cycle. The sine-cosine pair wraps the month onto a unit circle, so the model sees December and January as neighbours.

F. Weather shock binary flags

$$\text{rain\_deficit}(s,t) = \mathbf{1}[\text{rainfall}(s,t) < 50\,\text{mm}] \tag{10}$$
$$\text{rain\_excess}(s,t) = \mathbf{1}[\text{rainfall}(s,t) > 300\,\text{mm}] \tag{11}$$
$$\text{heat\_stress}(s,t) = \mathbf{1}[\text{temp\_mean}(s,t) > 38^\circ\text{C}] \tag{12}$$
$$\text{cold\_stress}(s,t) = \mathbf{1}[\text{temp\_mean}(s,t) < 5^\circ\text{C}] \tag{13}$$

The 50 mm deficit threshold is roughly half the long-run monthly average for the non-monsoon belt, around the agronomic point where vegetable crops show measurable water stress. At 300 mm excess, standing crops in low-lying fields begin to flood. The 38°C heat flag covers the temperature range known to cause fruit drop in tomatoes; the 5°C cold flag picks up north Indian winter months when cold damage affects rabi onion and tomato crops. Encoding these as binary flags rather than using the raw continuous value makes it easier for the model to fire a strong response at the threshold rather than trying to interpolate through it.

G. COVID lockdown flag

$$\text{covid\_lockdown}(t) = \mathbf{1}[t \in [\text{March 2020}, \text{June 2021}]] \tag{14}$$

This flag marks the lockdown period so that the model does not generalise from the broken supply chains and irregular trading of that period to normal months.

H. Final feature schema

Table IV groups the resulting features by type. The fusion signal xgb_log_pred is described in Section VI-D and is added only for the final model.

VI. METHODOLOGY

The pipeline has three stages. XGBoost takes the engineered features and outputs a point log-price estimate. TFT takes that estimate as one of its known future inputs and outputs three quantile forecasts per month: a lower bound, a median and an upper bound. Per-crop CQR then adjusts the interval on the 2022 calibration window so the target coverage holds on unseen data. Fig. 1 shows how the stages connect.

A. XGBoost baseline

XGBoost [19] is a gradient boosted decision tree ensemble that grows trees sequentially, each new tree trained to reduce the residual error of the previous ones. It is used as the point baseline because tree ensembles are effective on tabular features with lags, rolling means and weather columns, and are robust to outliers and missing values. The model is trained on

TABLE IV

FEATURES USED FOR MODELING, GROUPED BY TYPE. THE FUSION SIGNAL IS ADDED ONLY FOR THE FINAL MODEL.

Group	**Variables**
Static descriptors	commodity, market, state
Calendar / season	month, year, month_sin, month_cos, agricultural season
Price memory	price_lag_1m, price_lag_12m, rolling_3m, rolling_6m, yoy_change
Weather (continuous)	rainfall_monthly, temperature_mean, humidity_mean
Weather shock flags	rain_deficit, rain_excess, heat_stress, cold_stress
Event flag	covid_lockdown
Fusion signal	xgb_log_pred (final model only)

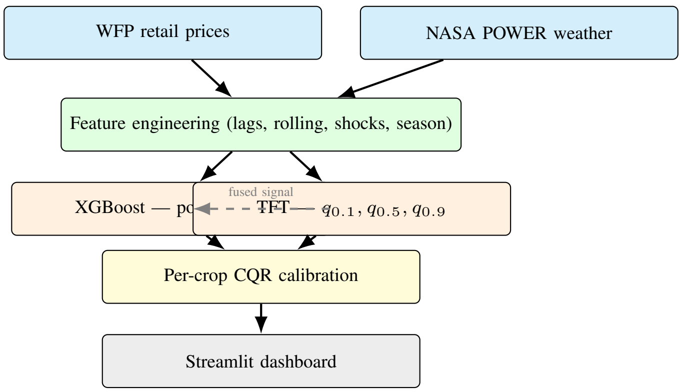

Fig. 1. Pipeline overview. WFP prices and NASA POWER weather feed shared feature engineering. XGBoost produces a point prediction that is injected into TFT as a known covariate (dashed arrow). TFT outputs three quantile predictions. Per-crop CQR adjusts the band. All artifacts load into the dashboard.

the engineered feature set (Table IV, without the fusion signal) to predict the log price for the next month.

Hyperparameters and rationale:

- learning_rate $= 0.05$: small enough that each tree only nudges the ensemble, large enough that 500 rounds is sufficient for convergence.
- max_depth $= 6$: captures interactions between weather, season and lag features without overfitting on this dataset size.
- n_estimators $=$ 500 with early stopping on validation MAE: stops adding trees once validation error stops improving.
- objective = `reg:squarederror`: standard squared-error regression loss for log price prediction.

B. Temporal Fusion Transformer (TFT)

TFT [12] is a deep neural network for multi-horizon forecasting. It takes a window of past months, known future values such as calendar position and seasonal flags, and static descriptors of the series, and produces a prediction for each future month of the horizon. Three components separate it from a plain LSTM: a Variable Selection Network that learns which inputs matter at each time step, a static enrichment layer that carries market-crop context through the whole network, and quantile output heads that produce three predictions per step rather than one.

Inputs. Static categorical inputs include crop type, market name and state, embedded into a vector shared across all time steps for that series. Time-varying known inputs are values whose future is available at prediction time: year, month, cyclically encoded month, agricultural season, the COVID lockdown flag, and in the final model the fused XGBoost prediction. Time-varying observed inputs are values whose future is unknown but past is observed: log price, rainfall, temperature, humidity, lag and rolling features, and weather shock flags.

Variable Selection Network (VSN). For each time step t, VSN computes a Softmax weight over all available inputs X_t given the static context c_s:

$$v_t = \mathrm{Softmax}(\mathrm{GRN}(X_t, c_s)) \tag{15}$$

GRN is a Gated Residual Network, a two-layer feed-forward block with a gated linear unit and a residual connection. The weight v_t indicates how much each input contributes at that time step and is exposed to the user as the explanation signal.

LSTM encoder-decoder. Past time steps pass through an LSTM encoder and the future horizon through an LSTM decoder, giving the model sequential context around the forecast.

Interpretable multi-head attention. A self-attention layer attends over the encoder output. TFT uses a simplified design that shares the value projection W_V across all attention heads, so head-averaged attention weights directly indicate which past months were most relevant for the current forecast.

Quantile heads. The decoder produces three predictions per future month: $\hat{q}_{0.1}$, $\hat{q}_{0.5}$ and $\hat{q}_{0.9}$. They are trained jointly using the pinball loss:

$$\rho_\tau(y, \hat{q}_\tau) = \begin{cases} \tau \cdot (y - \hat{q}_\tau), & y \geq \hat{q}_\tau \\ (1 - \tau) \cdot (\hat{q}_\tau - y), & y < \hat{q}_\tau \end{cases} \tag{16}$$

For $\tau = 0.5$ the pinball loss reduces to absolute error, giving the median prediction. For $\tau = 0.1$ it penalises over-prediction nine times more than under-prediction, which pushes the head to produce a low estimate. For $\tau = 0.9$ it does the opposite.

Hyperparameters were chosen to match the dataset size of 16,939 rows:

- hidden_size $=$ 32, hidden_continuous_size $=$ 16: kept small to avoid overfitting.

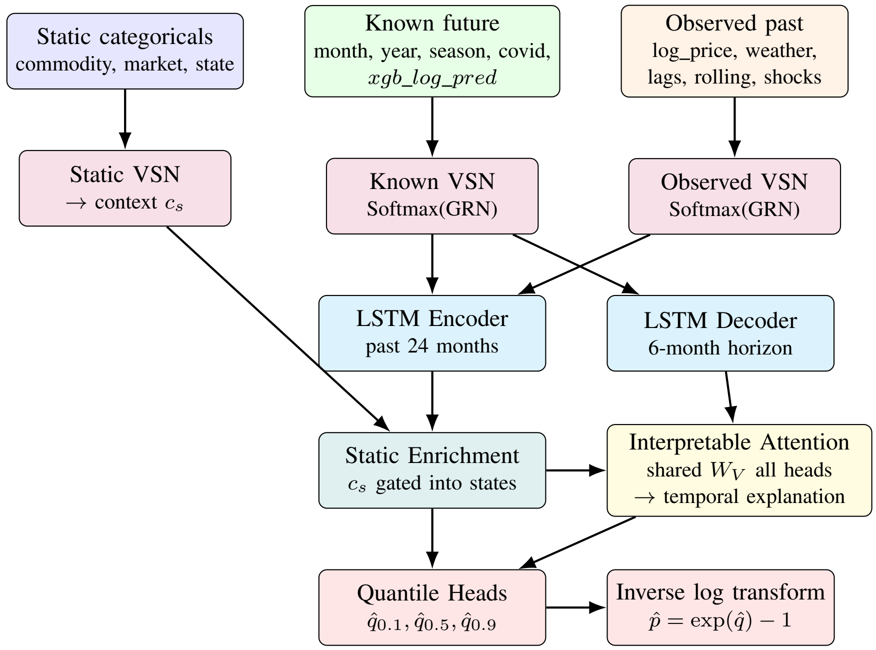

Fig. 2. TFT architecture. Three input streams — static categoricals (blue), known future covariates (green), and observed historical values (orange) — each pass through a Variable Selection Network (VSN, purple) that computes per-timestep input weights via a Softmax-weighted Gated Residual Network (GRN). Weighted inputs feed an LSTM encoder (past 24 months) and decoder (6-month horizon). A static enrichment gate injects market-crop context (c_s) into the LSTM states. An interpretable multi-head attention layer with shared value projection W_V attends over encoder history to produce temporal explanation weights. Three quantile heads output $\hat{q}_{0.1}, \hat{q}_{0.5}, \hat{q}_{0.9}$ per future step. An inverse log transform converts predictions back to price scale.

- lstm_layers $= 1$, attention_head_size $= 2$: minimal depth for monthly data.
- dropout $= 0.2$: standard regularisation.
- learning_rate $= 0.03$, batch_size $= 64$.
- max_encoder_length $= 24$, max_prediction_length $= 6$: two years of past context, six months ahead prediction.
- quantiles $= [0.1,\ 0.5,\ 0.9]$: central 80 percent prediction band plus a median.

C. Conformal Quantile Regression (CQR)

TFT produces a prediction band $[\hat{q}_{0.1},\ \hat{q}_{0.9}]$, but this band may be too narrow or too wide without adjustment. CQR [5] uses a calibration set to adjust the band so that it achieves the target coverage rate.

Step 1. Use the 2022 validation data as the calibration set.

Step 2. For each calibration example i, compute how far outside the band the actual log price y_i fell:

$$E_i = \max(\hat{q}_{0.1,i} - y_i,\quad y_i - \hat{q}_{0.9,i}) \tag{17}$$

E_i is positive when the actual price falls outside the band, and negative when the band is wider than needed.

Step 3. For each crop c, take the $(1-\alpha)$ quantile of all calibration scores. With $\alpha = 0.10$, this is the 90th percentile:

$$\Delta_c = Q_{1-\alpha}(\{E_i : i \in \mathcal{V}_c\}) \tag{18}$$

Step 4. Widen the band by Δ_c on both sides in log-price space, then convert back to price scale using Eq. 2:

$$\left[\hat{q}^{\text{cal}}_{0.1,i},\ \hat{q}^{\text{cal}}_{0.9,i}\right] = [\hat{q}_{0.1,i} - \Delta_c,\ \hat{q}_{0.9,i} + \Delta_c] \tag{19}$$

The median $\hat{q}_{0.5}$ is unchanged, so CQR improves band reliability without distorting the point prediction. Per-crop calibration is used because a single global offset would over-widen the stable rice band and under-widen the volatile tomato band. CQR provides finite-sample rather than exact coverage guarantees, so empirical coverage is reported from the test set rather than assumed from the target alpha.

D. XGBoost-Feature Fusion

The fused model is built by training a separate XGBoost model on data up to 2019 only, and using its predictions as an additional input to TFT. Let g_ϕ denote this auxiliary model. For each row with feature vector $\mathbf{z}_{s,t}$:

$$x^{xgb}_{s,t} = g_\phi(\mathbf{z}_{s,t}) \tag{20}$$

The value $x^{xgb}_{s,t}$ is appended to the time-varying known inputs of TFT. This is the only architectural change; everything else in TFT stays the same.

This is different from model stacking [4], where two models predict independently and their outputs are combined afterward. Here TFT sees the XGBoost prediction as a feature and decides through its VSN whether to rely on it or not. For a stable rice market the XGBoost signal is usually reliable and TFT can put high weight on it. During a tomato spike, where XGBoost can underestimate because tree models do not extrapolate to never-seen price levels, TFT can downweight that signal and fall back on its own sequential reasoning. The pre-2020 cutoff ensures that the auxiliary XGBoost model has not seen any 2022 or 2023 data, so CQR calibration on the 2022 window remains valid.

E. Checkpoint ensemble

To reduce variance from random initialisation, three TFT checkpoints are saved during one training run at different epochs that each met the validation criterion. At inference, all three checkpoints predict three quantiles per row, and the results are averaged in log-price space:

$$\hat{q}_\tau^{\text{ens}}(s,t) = \frac{1}{K}\sum_{k=1}^{K} \hat{q}_\tau^{(k)}(s,t) \tag{21}$$

Averaging in log space corresponds to a geometric mean in price space, which is consistent with the log-target training objective.

VII. Results and Evaluation

A. Evaluation metrics

- **MAE** $= \frac{1}{N}\sum_i |y_i - \hat{y}_i|$: average absolute error in Rs per kg.
- **RMSE** $= \sqrt{\frac{1}{N}\sum_i (y_i - \hat{y}_i)^2}$: penalises large errors more than MAE.
- **MAPE** $= \frac{100}{N}\sum_i \left|\frac{y_i - \hat{y}_i}{y_i}\right|$: relative error in percent; comparable across crops at different price levels.
- **Coverage** $= \frac{1}{N}\sum_i \mathbf{1}\left[\hat{q}^{\text{cal}}_{0.1,i} \le y_i \le \hat{q}^{\text{cal}}_{0.9,i}\right]$: fraction of test points inside the calibrated band.
- **Band Width** $= \frac{1}{N}\sum_i (\hat{q}^{\text{cal}}_{0.9,i} - \hat{q}^{\text{cal}}_{0.1,i})$: average width of the band in Rs per kg.

B. XGBoost baseline

A standalone XGBoost model gives the strongest point accuracy on the 2023 test period: **MAE = 1.58 Rs per kg, MAPE = 2.8%**. Tree ensembles are effective at exploiting engineered tabular features and are robust to the outliers present in volatile crop price data. However, this model produces only a single value per prediction with no band, no confidence measure and no per-timestep explanation. For market monitoring where future price spikes need to be anticipated, point accuracy alone is not enough.

C. TFT baseline

A plain TFT with no fusion and no calibration gives a weaker point result: **MAE = 10.53 Rs per kg, MAPE = 29.1%**. Its raw quantile band covers only 64.9 percent of test observations with an average width of 23.62 Rs per kg. A 65 percent coverage rate on a band meant to hold 80 percent of observations is not useful for procurement decisions, and the 29 percent MAPE is also too large for onions and tomatoes where a 10 percent error already translates to several rupees per kg at peak prices.

D. Progressive model improvements

Two changes were then applied on top of the plain TFT to reach the final model. Table V summarises how the metrics evolve.

1) **TFT with quantile forecasting (TFT-Base).** Plain TFT trained with pinball loss on three quantile heads. MAPE = 29.1%, coverage = 64.9%. The band is too narrow and the point accuracy is weak.
2) **XGBoost prediction fused as a known TFT input.** The auxiliary XGBoost prediction is injected as described in Section VI-D, and per-crop CQR is then applied on the 2022 window in log-price space. **MAE = 5.27 Rs per kg, MAPE = 11.4%, Coverage = 84.0%, Band Width = 11.99 Rs per kg.** The band is now both calibrated and narrow enough to be useful in practice.

E. Per-crop performance

Table VI reports the final model by crop. Onion accuracy is the best of the three. Onion prices follow regular kharif and rabi cycles, which the lag and rolling features pick up well. Tomatoes are the hardest. Tomato prices can double in a single month due to a local heat event or a transport disruption. The monthly weather inputs are too coarse to capture a one-week heat shock. Rice sits in between. Rice prices are stable, since government procurement keeps the floor steady.

TABLE VI
Crop-wise test performance for TFT-XGBFusion-CQR on the 2023 onward test period.

Crop	MAE (Rs per kg)	MAPE	Coverage
Onions	2.25	9.1%	90.9%
Tomatoes	8.13	13.0%	82.5%
Rice	4.81	11.7%	79.1%

Figure 3 compares per-crop MAE for the plain TFT and for the final fused-and-calibrated model, on the 2023 test period.

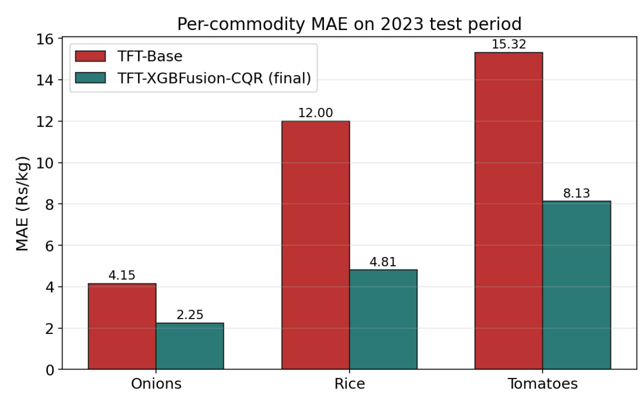

Fig. 3. Per-crop MAE on the 2023 test period: plain TFT vs the final fused-and-calibrated model.

Figure 4 plots the calibrated prediction band over the actual prices for each crop during the test period. The shaded region is the $q_{0.1}$ to $q_{0.9}$ calibrated band. The solid line is the median

TABLE V

TEST PERFORMANCE ON THE 2023 ONWARD PERIOD. COVERAGE AND BAND WIDTH APPLY ONLY TO PROBABILISTIC MODELS. THE FUSED AND CALIBRATED MODEL IS THE FINAL ONE.

Model	MAE (Rs per kg)	MAPE	Coverage	Band Width (Rs per kg)
XGBoost (point baseline)	1.58	2.8%	–	–
TFT-Base	10.53	29.1%	64.9%	23.62
TFT-XGBFusion-CQR	**5.27**	**11.4%**	**84.0%**	**11.99**

forecast. The onion and rice plots track the actual price closely. For tomatoes, the band correctly widens during the mid-2023 price surge, which matches the higher uncertainty in that period.

F. Statistical validation of the improvement

Three statistical tests were run on the 792 test rows to confirm that the gain over the plain TFT is not due to chance. The same rows are used for both models, so the comparison is fully paired.

Paired two-sided t-test. The null hypothesis H_0 is that the mean absolute-error difference between the two models is zero. Result: mean MAE reduction = 5.577 Rs per kg, $t = -22.34$, $p = 4.6 \times 10^{-86}$. H_0 is rejected. The gain is statistically real.

Wilcoxon signed-rank test. The t-test assumes the error differences are roughly normal. The Wilcoxon test does not. It works on the ranks of the differences and acts as a non-parametric backup. Result: $p = 1.2 \times 10^{-95}$. Both tests agree.

Cohen's d effect size. At large n, a tiny improvement can still be statistically significant. Cohen's d measures the standardised gap between the two error distributions. A value above $|d| = 0.5$ counts as large. Result: $d = -0.79$. The gain is large in practical terms, not just statistical.

Table VII summarises the three tests for the key comparison.

TABLE VII

PAIRED ABLATION TESTS ON $n = 792$ MATCHED MARKET-DATE ROWS FROM THE 2023 TEST PERIOD. COHEN'S $|d| > 0.5$ INDICATES A LARGE PRACTICAL EFFECT.

Statistic	Fusion vs TFT-Base
ΔMAE (Rs per kg)	-5.577
t	-22.34
p (paired t)	4.6×10^{-86}
Wilcoxon p	1.2×10^{-95}
Cohen's d	-0.79

Empirical coverage binomial test. For each crop, an exact two-sided binomial test checks whether the count of test points inside the calibrated band is consistent with the 80 percent nominal target. Table VIII lists the results. Rice ($p = 0.695$) and tomatoes ($p = 0.310$) match the 80 percent target. Onions over-cover at 90.9 percent ($p < 0.001$). The band is therefore wider than needed for onions. Given the onion price spike history, a wider band is acceptable.

TABLE VIII

PER-CROP BINOMIAL TEST OF EMPIRICAL COVERAGE. H_0: EMPIRICAL COVERAGE = 80% (NOMINAL). $p > 0.05$ MEANS STATISTICALLY CONSISTENT WITH NOMINAL.

Crop	n	Hits	Coverage	p
Onions	242	220	90.9%	< 0.001
Tomatoes	297	245	82.5%	0.310
Rice	253	200	79.1%	0.695
All	**792**	**665**	**84.0%**	**0.005**

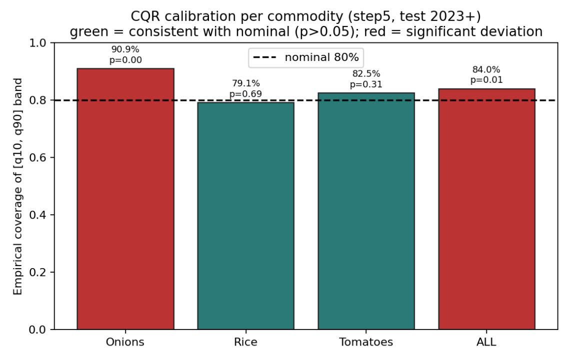

Fig. 5. Empirical coverage of the calibrated 80 percent prediction band by crop on the 2023 test set. The dashed line is the nominal 80 percent target. Bars statistically consistent with nominal ($p > 0.05$) are shown in green.

VSN bootstrap stability. An explanation is only useful if it is reproducible. To check the stability of TFT's feature ranking, 1000 random resamples of market-crop series are drawn with replacement. Each resample gives a feature ranking from the VSN encoder weights. Pairwise Kendall rank correlations are then computed across all resampled rankings. A τ value close to 1 means the rankings are nearly identical across resamples. A value close to 0 means the rankings are random. Result: **mean** $\tau = 0.942$**, 95 percent confidence interval** $[0.895,\ 0.990]$. The ranking is highly stable.

The top features fall into three groups. Weather sits at the top: mean temperature (0.143), excess rainfall (0.143) and deficit rainfall (0.140), followed by humidity (0.066). Calendar and event variables come next: year (0.060), the COVID lockdown flag (0.059) and the agricultural season (0.042). Price-memory features round out the top ten: log price (0.041), the six-month rolling mean (0.038) and the twelve-month price lag (0.038). The ordering matches what a domain expert would expect. Weather drives supply. The season and event flags handle structural shifts. Price memory carries the short-term momentum.

Fig. 4. Calibrated TFT-XGBFusion-CQR prediction band (shaded) and median forecast (solid line) over actual prices for onions (top), tomatoes (middle) and rice (bottom) on the 2023 test set.

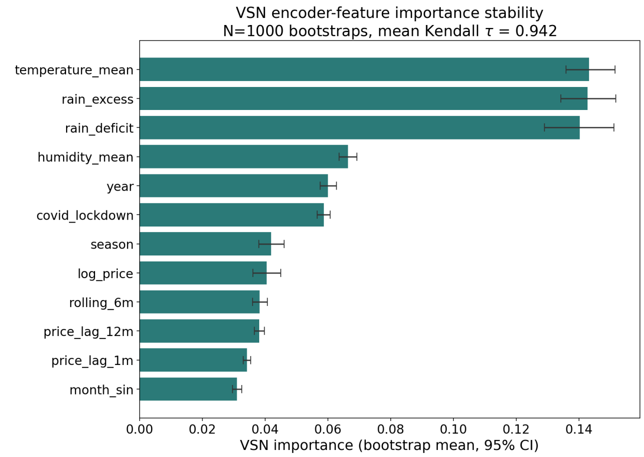

Fig. 6. Top TFT encoder features ranked by bootstrap-mean VSN importance with 95 percent confidence intervals (1000 resamples). Kendall $\tau = 0.942$ confirms the ranking is stable across resamples.

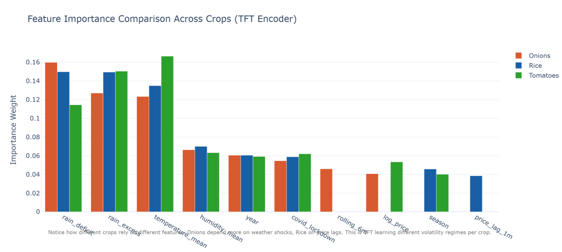

Fig. 7. Comparison of encoder and decoder VSN feature importances. Weather and price-memory features dominate consistently across both the encoder and decoder pathways.

A high VSN weight on rain deficit means the trained model relies on that variable heavily during prediction. It does not prove that a rainfall deficit caused the price move. The point is auditability. The attribution patterns are stable and domain-consistent, so each forecast can be checked by a domain expert.

G. Comparison with prior work

Table IX gives an indicative comparison with prior Indian crop price forecasting studies. Direct numerical comparison is difficult because each study uses different crops, markets, time periods and prediction frequencies. Monthly data smooths within-month spikes that daily or weekly models must absorb, so monthly MAPE values are naturally lower than daily MAPE values.

XGBoost on its own gives the lowest point MAPE because tree ensembles are very effective on engineered tabular features. It does not produce a calibrated band or per-timestep explanation. The fused and calibrated model gives the strongest combined result: a competitive MAPE of 11.4 percent, an empirically valid band at 84 percent coverage with an average width of around 12 Rs per kg, and a feature ranking that is stable across 1000 bootstrap resamples. For market monitoring, where an overconfident or underconfident forecast can lead to poor procurement or storage calls, this combined behaviour is more valuable than the lowest point error alone.

VIII. Implementation and Dashboard

The trained artifacts — the XGBoost model file, the TFT checkpoint and the per-crop CQR offsets — are loaded into a Streamlit application. The application has four views for any chosen crop and market. The active model is selected at startup based on the artifacts present, so the dashboard always runs the best available model.

Figure 8 shows the main view, which is the one used for actual decision making. For each month from one to six ahead, the view shows the median predicted price, the lower and upper bounds of the calibrated band, and a risk label. The risk label is derived from the relative band width and takes three values: stable, uncertain or high risk. Two months may have similar median forecasts but very different bands. That difference matters for a trader or a procurement planner. The other three views support the same workflow. A price-tracking view overlays the actual price with XGBoost, the TFT median and the calibrated band. An explainability view shows XGBoost gain values, TFT VSN weights and temporal attention. A test-validation view overlays the actual prices on the 2023 test period so the model can be audited on data it never saw. The artifacts are loaded once at startup. The historical panel is read from processed CSV files. This keeps the prototype lightweight and self-contained.

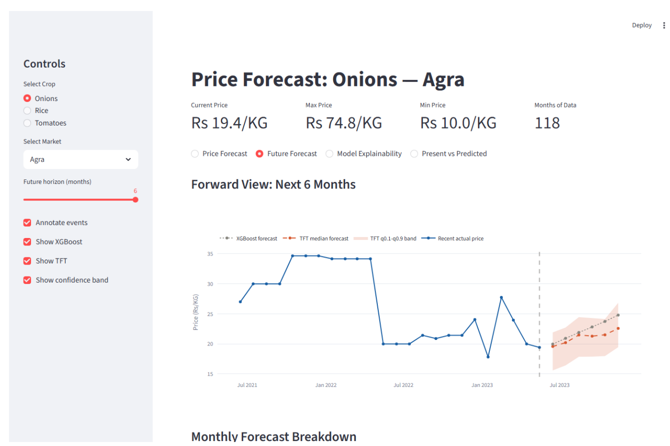

Fig. 8. Future forecast view of the dashboard. For each of the next one to six months the system shows the median predicted price, the calibrated band, and a risk label derived from the relative band width.

IX. Conclusion and Future Work

This paper has presented a crop price forecasting setup that combines a gradient boosted point predictor (XGBoost), a probabilistic deep model with built-in interpretability (TFT), feature fusion between the two, and per-crop CQR calibration. On the 2023 test period for onions, tomatoes and rice across 123 market-crop series, the final model reaches 11.4 percent MAPE and 84.0 percent empirical coverage for the calibrated 80 percent band, with an average width of around 12 Rs per

TABLE IX
INDICATIVE COMPARISON WITH PRIOR INDIAN CROP PRICE FORECASTING WORK. MAPE VALUES ARE NOT DIRECTLY COMPARABLE ACROSS STUDIES DUE TO DIFFERENCES IN CROPS, MARKETS AND UPDATE FREQUENCIES.

Work	**Crops**	**MAPE**	**Reports uncertainty?**
Nayak et al. [1]	Tomato, Onion, Potato	8–15%	no
Manogna et al. [2]	Multiple crops	5–20%	no
Paul et al. [3]	Brinjal	7–12%	no
This work	Onions, Tomatoes, Rice	**11.4%**	**yes (84% coverage)**

kg. Three statistical tests back this result: a paired t-test gives $p = 4.6 \times 10^{-86}$, a Wilcoxon test gives $p = 1.2 \times 10^{-95}$, and Cohen's $d = -0.79$ indicates a large practical effect size. The feature ranking is stable across 1000 bootstrap resamples (Kendall $\tau = 0.942$).

One trade-off is worth stating directly. XGBoost on its own achieves a lower point MAPE (2.8 percent) than the fused probabilistic model, but it produces no band, no confidence measure and no per-timestep explanation. The contribution of this work is the combined system that produces all three, exposed through a usable dashboard.

The data used is monthly, so weekly and daily price shocks are not directly observable. Weather inputs are monthly satellite means, so a one-week heat shock can be invisible to the model. Three crops are studied, and the same approach will need re-tuning for other commodities. The test period covers roughly seven months. The dashboard is a prototype and does not yet refresh data automatically.

The biggest gap is the time granularity. Monthly data smooths over short heat or rain events that can push tomato prices up by half within a single retail cycle. Switching to weekly mandi data from Agmarknet would let the model pick up these shorter shocks. On the feature side, news headlines and commodity sentiment scores could flag supply disruptions before the price series moves. The current input set is weather-only and cannot do this. The CQR offsets are computed once on the 2022 calibration window. If the underlying price regime shifts, the coverage will drift. A periodic recalibration step tied to incoming WFP data would handle this. Finally, the system covers three crops. Extending it to pulses, potato and edible oils would test whether the fusion-plus-calibration approach holds for commodities with different volatility and different policy support.

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